

```

cp.queue <- db
cp.currentSize++
cp.isNotEmpty.Signal()

return nil
}

```

This function, *Add()*, adds a connection to the pool. It first checks whether the connection is 'nil' and returns an error if it is. Then, it acquires the lock to ensure exclusive access to the shared state of the connection pool. If the pool is currently full, the function waits until a connection is released from the pool.

Once there is space in the pool, the function enqueues the connection onto the queue, increments *currentSize*, and signals that the pool is not empty. The function returns 'nil' to indicate success.

Closing the connection pool

As the name suggests, we will implement a *Close* function which will be responsible for closing all database connections in the pool. It starts by acquiring a lock and then iterates through all the connections in the pool, and closes them one by one. After closing each connection, it decrements the *currentSize* counter and signals any waiting goroutines that there is space available in the pool now.

```

func (cp *ConnectionPool) Close() {
    cp.lock.Lock()
    defer cp.lock.Unlock()

    for cp.currentSize > 0 {
        db := <-cp.queue
        db.Close()
        cp.currentSize--
        cp.isNotFull.Signal()
    }

    close(cp.queue)
}

```

With the basic building blocks in place, can we use whatever we have as of now and run our own connection pool? Not yet. We still need a way to initialise our connection pool. What's missing you ask? Answers to that and more will be covered in the next part of this series. Watch out for this space! 

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